

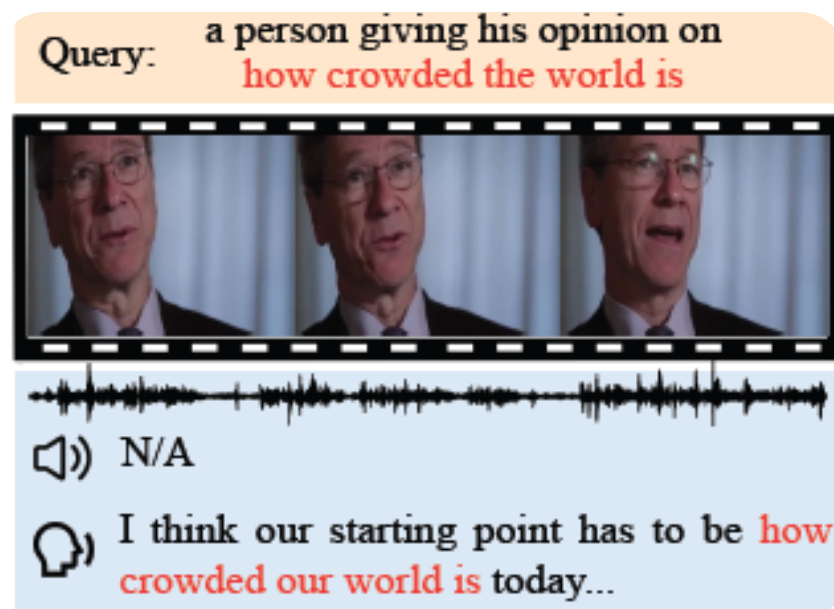
Query-Aware Gate Residual Fusion in Text-Video Retrieval

Query에 따른 Audio/Speech feature gate weight 조절

Query-Aware Gate Residual Fusion in T2VR



Audio-related



Speech-related

T2VR (Text-to-Video Retrieval)

- Aims to retrieve the video that best matches with a given text query

Most existing methods rely mainly on visual information processing.

However, some real-world queries require also audio/speech information.

Recent studies:

- Visual + Audio (AVIGATE, 2025)
- Visual + Audio + Speech (SAVE, 2026)

However, should we always use audio and speech?



Query 1: "Basketball players making a shot"

→ Visual Is all you need.

Query 2: "This is last shot in the last seven seconds"

→ Should focus on speech text

Unnecessary audio / speech information may give damage
in retrieval performance for many queries.

Proposed Method:

Determine how much audio / speech information should be **weighted** in retrieval

(based on query)

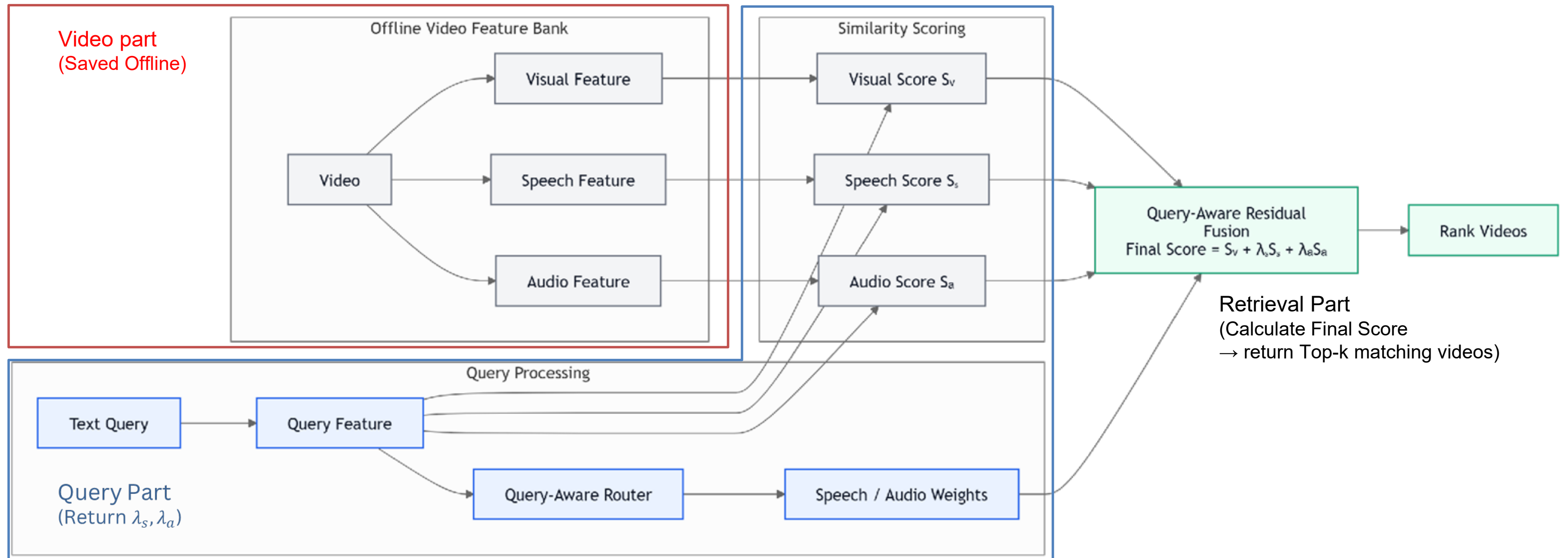
Query 1: "Basketball players making a shot"

Query 2: "This is last shot in the last seven seconds"

Score = Visual Similarity Score

Score = Visual Similarity Score + 0.5 Speech Similarity Score

Final Score = Visual Score + λ_a Audio Score + λ_s Speech Score

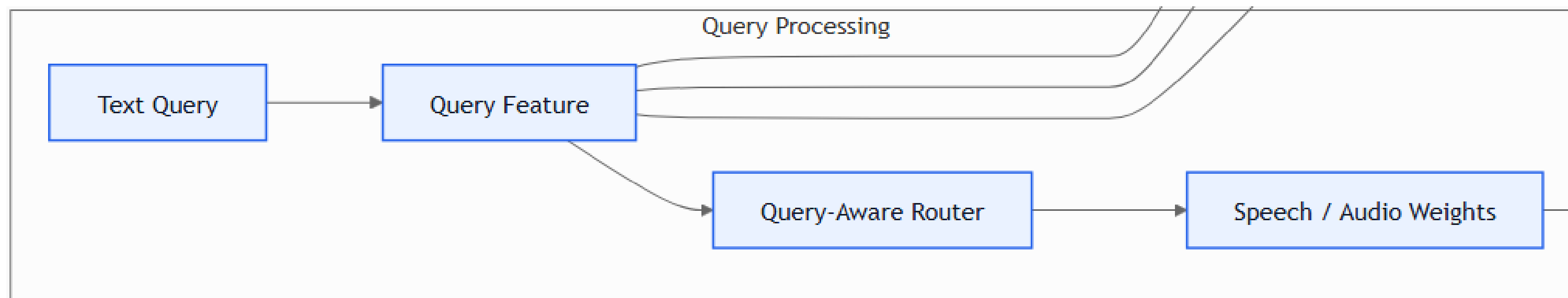


Visual / Speech Encoder: CLIP (ViT/B-32) + Transformer

Audio Encoder: AST + Projection

Speech Encoder: Whisper + CLIP

Query Encoder: CLIP



Query-Aware Router: 2-layer $MLP(q) \rightarrow \lambda_a, \lambda_s$

$$\mathcal{L} = \mathcal{L}_{\text{NCE}} + \mathcal{L}_{\text{sparsity}}$$

$$\mathcal{L}_{\text{NCE}} = \text{InfoNCE}(S_V + \lambda_s S_S + \lambda_a S_A)$$

$$\mathcal{L}_{\text{sparsity}} = w_{\text{sparsity}} \cdot \text{mean}(\lambda_s, \lambda_a)$$

- **NCE Loss**
Encourages higher λ_a, λ_s when they improve positive query-video matching
- **Sparsity Loss**
Prevents the model from using speech/audio information when it is unnecessary

Result

Test set: MSR-VTT-7k (2,990 Video/ 59,794 Query)

Visual Only

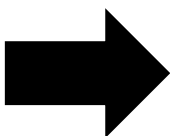
R@1: 22.03

R@5: 44.23

R@10: 54.62

MedianRank: 8.00

MeanRank: 74.97



Visual + Audio/Speech

R@1: 23.44

R@5: 46.11

R@10: 56.56

MedianRank: 7.00

MeanRank: 69.68

Residual Fusion Log

[51165] video7360 | **rank V/S/A/R=9/2/267/4**

query: a woman in a green patterned shirt is describing the ingredients going into her shrimp etouffee

[53043] video7577 | **rank V/S/A/R=7/6/56/1**

query: a women in blue striped grey shirt is explaining about the planets

[44993] video7072 **rank V/S/A/R=11/27/8/6**

query: a woman in a purple shirt talking into her computer about a course in college writing

Ablation 1: Auxiliary vs. Fusion Residual

visual information이 아예 필요 없는 query도 존재
→ λ_s 와 λ_a 이 커지면 visual 정보 반영 비율이 감소하도록 (Fusion)

	Speech Cap = 0.4, Audio Cap = 0.2	Cap X
Baseline	22.03 / 44.23 / 54.62	
Auxiliary (Visual score + Gated Speech/Audio)	23.44 / 46.11 / 56.56	23.60 / 46.06 / 56.52
Fusion (Fusion of Visual/Speech/Audio)	22.08 / 44.00 / 54.17	21.88 / 43.28 / 53.48

Ablation 2: Speech/Audio Cap

Metrics: Recall@1 / Recall@5 / Recall@10

Audio cap Speech Cap	0.1	0.2	0.3
0.2	23.30 / 45.81 / 56.22	23.31 / 45.86 / 56.26	23.34 / 45.87 / 56.31
0.3	23.40 / 46.08 / 56.46	23.34 / 46.03 / 56.37	23.38 / 46.04 / 56.45
0.4	23.41 / 46.09 / 56.50	23.44 / 46.11 / 56.56	23.41 / 46.14 / 56.51
0.5	23.38 / 45.97 / 56.43	23.48 / 46.07 / 56.42	23.45 / 46.09 / 56.35